

NATIONAL INSTITUTE OF ENGINEERING

END SEMESTER EXAMINATION - MAY 2027

Duration: 3 Hours

Max Marks: 80

Instructions: Attempt any four questions. Figures to the right indicate full marks.

Q1. Describe the scheduling policy and derive its average waiting time. [10]

Q2. Differentiate between the two normalisation forms with a worked example. [10]

Q3. Discuss the error control mechanism and analyse its overhead. [10]

Q4. Derive the expression for throughput and state all assumptions clearly. [10]

Q5. Compare the two indexing structures and evaluate their lookup complexity. [10]

Q6. Explain the deadlock avoidance algorithm and apply it to a sample matrix. [10]

Q7. Illustrate the encoding scheme and compute the resulting bandwidth. [10]